Risk Management Document

PetroPulse - Real-Time Sentimental Analytics Dashboard

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1 Introduction

Risk management is an essential process in software engineering that helps identify potential issues before they impact the system. PetroPulse relies on real-time data ingestion and machine learning models, making it susceptible to technical, operational, and security risks. Proper risk handling ensures uninterrupted analytics, system security, and user satisfaction.

2 Objectives

- Identify possible risks in the system development and deploment
- Analyze the probability and impact of each risk
- Defining mitigation strategies
- Improve system reliability and security

3 Risk Identification

3.1 Technical Risks

- API failures due to rate limits or downtime
- Model inaccuracies affecting sentiment results
- Database failures or data loss

3.2 Security Risks

- Unauthorized access to system
- Leakage of API keys
- Data breaches and privacy issues

3.3 Functional Risks

- Incorrect sentiment classification
- Errors in dashboard visualisation
- Data inconsistancy

3.4 Operational Risks

- Network failures
- Server downtime

3.5 Team Risks

- Lack of coordination among developers
- Merge conflicts in version control

4 Risk Analysis

Risk	Probability	Impact	Level
API failure	Medium	High	High
Model Inaccuracy	Medium	Medium	Medium
Data breach	Low	Very High	High
Server Downtime	Medium	High	High
Merge Conflict	Medium	Low	Medium

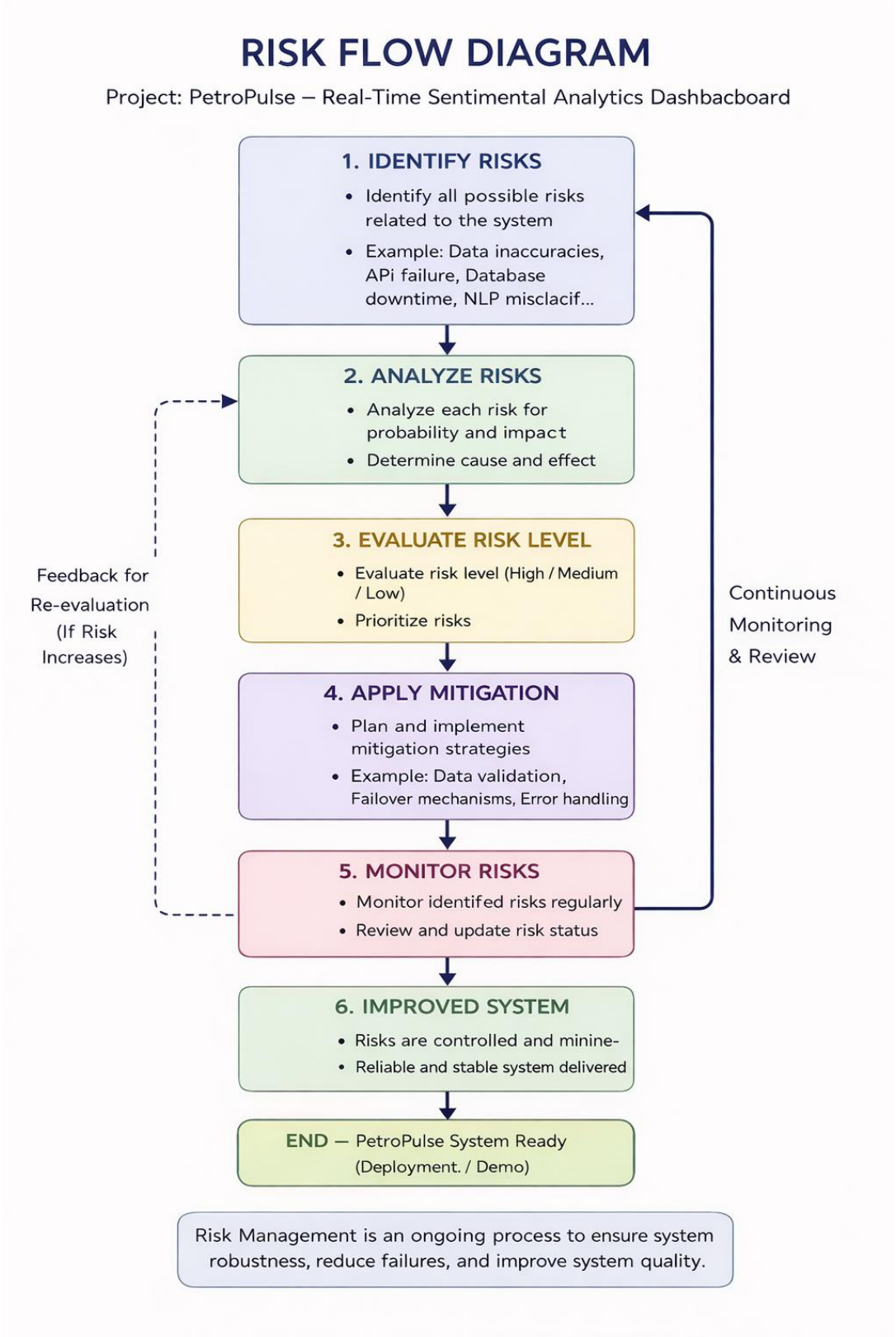
5 Risk Mitigation Strategies

Risk	Mitigation
API failures	Retry mechanisms and fallback APIs
Model Issues	Continuous training and validation datasets
Security issues	Encryption and authentication
Server Issues	Cloud backup and monitoring tools
Development issues	Follow structured Git workflows

6 Risk Monitoring

- Monitor logs and system performance
- Conduct regular testing
- Track API health
- Reviewing Git commits and changes

7 Risk Flow Diagram



8 Conclusion

Risk management is essential to maintaining the stability, security, and performance of PetroPulse. By proactively identifying and addressing vulnerabilities, the platform minimizes downtime, prevents data breaches, and ensures analytical accuracy. This continuous risk mitigation framework empowers PetroPulse to operate with resilience and consistency in the demanding energy sector.